

Quiz 8

Hardest...

- direct isometries: $\boxed{B}$ glide rotations
 $\boxed{D}$ translations

- $\Phi(x) = Ax + b$ indirect isometry of $\mathbb{E}^3$.

$$\Rightarrow A \in O(3), \det A = -1.$$

$\boxed{B}$

$\boxed{A}$. A admits -1 as eigenvalue

- $\mathbb{E}^3$, $P(3,2,0)$, $v(0,1,1)$, $\bar{u}: y-1=0 \Rightarrow y=1$.

$\text{Ref}_{\bar{u}}^\perp \rightarrow$ orthogonal reflection in $\bar{u} \Rightarrow \det = -1$ (indirect isometry)

$\text{Ref}_{\bar{u},v} \rightarrow$ parallel reflection in $\bar{u}$ along $v \Rightarrow \det = -1$ $\uparrow$

$$\Phi \rightarrow \cancel{\text{Ref}_{\bar{u},v} \circ \text{Ref}_{\bar{u}}^\perp} \quad \text{Ref}_{\bar{u}}^\perp \circ \text{Ref}_{\bar{u},v}$$

- $\Phi(P) = ?$

$\rightarrow$ line through P parallel to v : $P + t \cdot v = (3,2,0) + t \cdot (0,1,1) = (3,2+t,t)$

$$\mathcal{L} \cap \bar{u} \Rightarrow 2+t = y = 1 \Rightarrow t = -1$$

$$\Rightarrow (3,2-1,1) = (3,1,1) = H, \text{ midpoint}$$

$$\boxed{P' = 2H - P} = 2(3,1,1) - (3,2,0) = (6,2,2) - (3,2,0) = (3,0,2) \\ = \text{Ref}_{\bar{u},v}(P)$$

$\rightarrow$ orthogonal refl. in $\bar{u}: y=1$

$$\boxed{(x,y,z) \mapsto (x, 2-y, z)}$$

$$(3,0,-2) \mapsto (3,2,-2)$$

$$\Phi(P) = (3,2,-2)$$

$$4) \quad \Phi = \text{Ref}_{\bar{u}}^\perp \circ \text{Ref}_{\bar{u},v}$$

$$(-1) \cdot (-1) = 1 = \det \Rightarrow \text{direct isometry } \boxed{A}.$$

$$5) \quad \Phi \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix}$$

$\text{Ref}_{\pi, v}(x, y, z)$

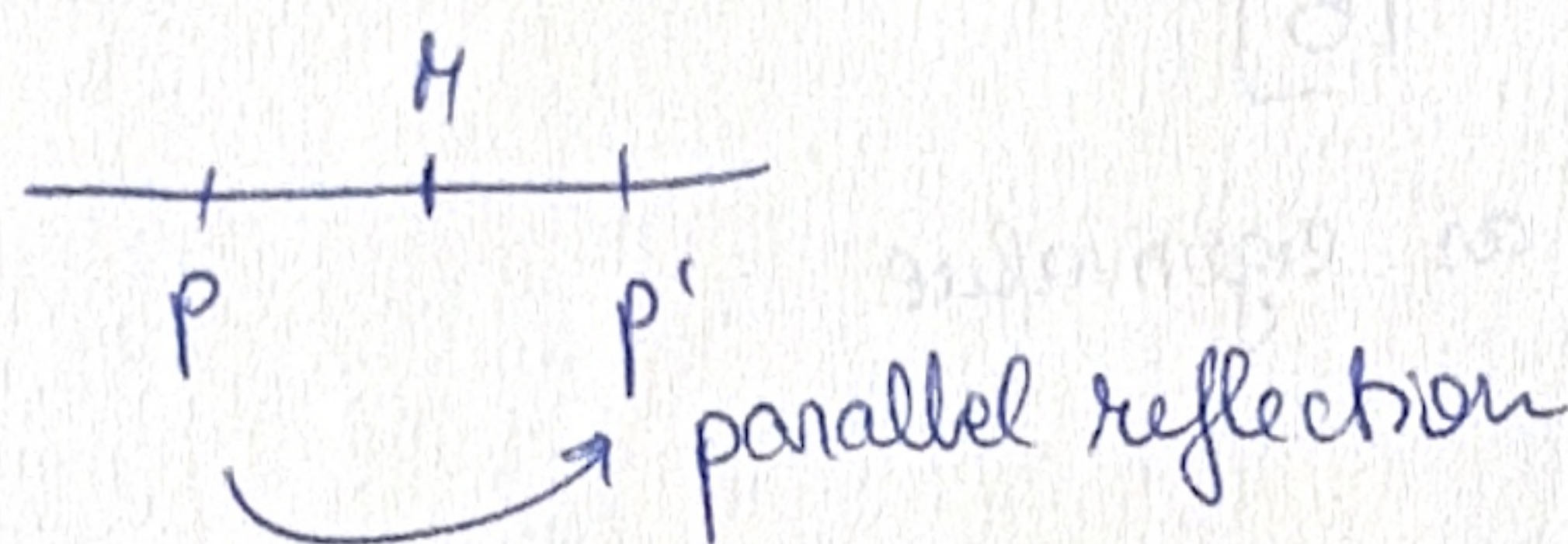
$P(x, y, z)$

$P + t \cdot (0, 1, 1) = (x, y+t, z+t)$ where it meets the plane.

$y+t=1 \Rightarrow t=1-y.$

$M(x, 1, z+1-y) \leftarrow$ intersection point with the plane.

$\hookrightarrow$ midpoint



$\boxed{P' = 2M - P} = (2x, 2, 2z+2-2y) - (x, y, z) = (x, 2-y, z-2y+2).$

$\text{Ref}_{\frac{\pi}{2}}$

$\text{Ref}_{\pi, v}(x, y, z)$

$(x, y, z) \mapsto (x, 2-y, z)$

....

$\boxed{9 = 45 = 9}$

$(5, 1-5, 2) \mapsto (5, 4, 2)$

$(5, 5, 2) \mapsto (5, 0, 2)$

$(5, 5, 2) \mapsto (9, 1)$

$(12, 0, 0) \mapsto 12$

$12 = 12$

$(12, 1, 1) \mapsto 12$

$$\mathbb{E}^3, P(1,2,1), v(0,1,1), \pi: z=1=0$$

$\text{Ref}_{\pi, v}$ in π along $v \rightarrow$ not isometry (parallel refl.) $\Rightarrow$ not isom.

$\text{Rot}_{Ox, 45^\circ}$ around (x) -axis $\rightarrow$ isometry

$$\phi = \text{Ref}_{\pi, v} \circ \text{Rot}_{Ox, 45^\circ}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}}X - \frac{1}{\sqrt{2}}Y \\ \frac{1}{\sqrt{2}}X + \frac{1}{\sqrt{2}}Y \end{pmatrix}$$

$$\Rightarrow x = \frac{X-Y}{\sqrt{2}} \quad (1) \quad \Rightarrow y = \frac{Y-Z}{\sqrt{2}} \quad (1)$$

$$\Rightarrow y = \frac{X+Y}{\sqrt{2}} \quad (2) \quad z = \frac{Y+Z}{\sqrt{2}} \quad (2)$$

$$(1) + (2) \Rightarrow x + y = \frac{X-Y+X+Y}{\sqrt{2}} = \frac{2X}{\sqrt{2}} = \sqrt{2}X \Rightarrow X = \frac{x+y}{\sqrt{2}}$$

$$(2) - (1) \Rightarrow y - x = \frac{X+Y-X+Y}{\sqrt{2}} = \frac{2Y}{\sqrt{2}} = \sqrt{2}Y \Rightarrow Y = \frac{y-x}{\sqrt{2}}$$

$$(1) + (2) \Rightarrow y + z = \sqrt{2}Y \Rightarrow Y = \frac{y+z}{\sqrt{2}}$$

$$(2) - (1) \Rightarrow \dots \quad z = \frac{y+z}{\sqrt{2}}$$

$$\text{Rot}_{Ox, 45^\circ}(x, y, z) = \left(x, \frac{y-z}{\sqrt{2}}, \frac{y+z}{\sqrt{2}}\right) = \left(1, \frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right) = Q$$

$$P(1,2,1) \Rightarrow X=1$$



$$Y = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$Z = \frac{2+1}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

$$P' = P + t \cdot v = (1, 2, 1) + t \cdot (0, 1, 1)$$

$$\text{line } \cap \text{ plane } \begin{cases} \ell(t) = Q + t \cdot v = \left(1, \frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right) + t \cdot (0, 1, 1) = \left(1, \frac{1}{\sqrt{2}} + t, \frac{3}{\sqrt{2}} + t\right) \\ z=1 \end{cases}$$

$$\Rightarrow \frac{3}{\sqrt{2}} + t = 1 \Rightarrow t = 1 - \frac{3}{\sqrt{2}}$$

$$\ell(t) = \left(1, \frac{1}{\sqrt{2}} + 1 - \frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}} + 1 - \frac{3}{\sqrt{2}}\right) = \left(1, 1 - \frac{2}{\sqrt{2}}, 1\right) = H \in \pi$$

$$\boxed{Q' = 2H - Q} = 2\left(1, 1 - \frac{2}{\sqrt{2}}, 1\right) - \left(1, \frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right) = \left(1, 2 - \frac{5}{\sqrt{2}}, 2 - \frac{3}{\sqrt{2}}\right)$$

$$= \text{Ref}_{\pi, v}(Q)$$

7) Φ not isom. $\boxed{e}$

$$8) \Phi\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix}$$

$$\Phi = \text{Ref}_{\Pi, v} \circ \text{Rot}_{Ox, 45^\circ}$$

$$\text{Rot}_{Ox, 45^\circ}(x, y, z) = \left(x, \frac{y-z}{\sqrt{2}}, \frac{y+z}{\sqrt{2}}\right)$$

$\text{Ref}_{\Pi, v}$ along $v = (0, 1, 1)$, in $\Pi: z=1$: $(x, y, z) \mapsto (x, y+2-2z, 2-z)$

$$\Phi(x, y, z) = \left(x, \frac{y-z}{\sqrt{2}} + 2 - 2 \cdot \frac{y+z}{\sqrt{2}}, 2 - \frac{y+z}{\sqrt{2}}\right)$$

$$\Phi\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{3}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$

9) $\mathbb{E}^3$, orthogonal reflection:

$$R\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \frac{1}{9} \begin{bmatrix} 7 & -4 & 4 \\ -4 & 1 & 8 \\ 4 & 8 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Equation of the reflection plane of the reflection R .

$$\boxed{R = I - 2m \cdot m^T} \text{ for an orthog. refl., } m = \text{unit normal vector}$$

? $m = \frac{1}{3}(1, 2, -2) \Rightarrow (1, 2, -2) \text{ a normal vector.}$

$\Rightarrow$ refl. plane: $x + 2y - 2z = 0$.

Quiz 8 - redone.

1. dir. isometries? $\boxed{B}$, $\boxed{D}$

2. $\Phi = Ax + b$ indir. isom.

$\boxed{A} \rightarrow -1$ eigenval.

$\boxed{B}$ $\det A = -1 \Rightarrow \in O(3)$.

I. $\mathbb{H}^3$, $P(3,2,0)$, $u(0,1,1)$, $\bar{u}: y-1=0 \Rightarrow a=(0,1,0)$, $a_{n+1}=-1$.

$$\Phi = \text{Ref}_{\frac{1}{u}} \circ \text{Ref}_{\bar{u}, u}$$

3) $\Phi(P) = ?$

gain. $\text{Ref}_{\frac{1}{u}} = P - 2 \cdot \frac{a^T P + a_{n+1} \cdot a}{|a|^2} \cdot a = (3,2,0) - 2 \cdot \frac{1}{1} \cdot (0,1,0) = (3,2,0) - (0,2,0) = (3,0,0)$

$$a^T P + a_{n+1} = (0 \ 1 \ 0) \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} - 1 = 2 - 1 = 1.$$

$$|a|^2 = (\sqrt{1^2})^2 = 1.$$

$$\text{Ref}_{\frac{1}{u}}(P) = \left(I_3 - 2 \cdot \frac{a \otimes a}{|a|^2} \right) P - 2 \cdot \frac{a_{n+1}}{|a|^2} \cdot a$$

$$\text{Ref}_{\frac{1}{u}, u} = \left(I_3 - 2 \cdot \frac{u \cdot a^T}{u^T \cdot a} \right) P - 2 \cdot \frac{a_{n+1}}{u^T \cdot a} u = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} - 2 \cdot \frac{-1}{1} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\hookrightarrow \text{lin. part} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$u \cdot a^T = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$a \otimes a = a \cdot a^T = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3x3 3x1

$$u^T \cdot a = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$$

3x3 x 3x1

$$\Phi(P) = \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right) x - 2 \cdot \frac{-1}{1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\Phi \left(\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} =$$

$$\hookrightarrow = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

4) ~~1A~~

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\rightarrow AA^T = I_3?$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & -2 & 5 \end{bmatrix} \neq I_3$$

$\Rightarrow$ not isom.

5). AGAIN.

$$\mathbb{E}^3, P(1,2,1), v(0,1,1), \pi: z-1=0, a=(0,0,1), a_{n+1}=-1.$$

$$\phi = \text{Ref}_{\pi, v} \circ \text{Rot}_{Ox, 45^\circ}$$

$$\text{Rot}_{Ox, 45^\circ} = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} X \quad \mathbb{E}^3$$

$$\Rightarrow \text{Rot}_{Ox, 45^\circ} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 45 & -\sin 45 \\ 0 & \sin 45 & \cos 45 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} =$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{Ref}_{\pi, v} = \left(I_m - 2 \cdot \frac{v \cdot a^T}{v^T a} \right) X - 2 \cdot \frac{a_{n+1}}{v^T a} v = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & -1 \end{bmatrix} X + 2 \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$v \cdot a^T = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} (0 \ 0 \ 1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$v^T a = (0 \ 1 \ 1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & -1 \end{bmatrix}$$

$$\Rightarrow \phi(X) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & -1 \end{bmatrix} \cdot \left(\frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} X \right) + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 1 \end{bmatrix} X + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

Calculus mistakes?

$$b) \phi(P) = \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} \\ 1 \\ -5 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} =$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} \\ 1 \\ -5 \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 2\sqrt{2} \\ 2\sqrt{2} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} \\ 1+2\sqrt{2} \\ -5+2\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1+2\sqrt{2}}{\sqrt{2}} \\ \frac{-5+2\sqrt{2}}{\sqrt{2}} \end{bmatrix} =$$

$$= \left(1, \frac{\sqrt{2}(1+2\sqrt{2})}{2}, \frac{\sqrt{2}(2\sqrt{2}-5)}{\sqrt{2}} \right) =$$

$$= \left(1, \frac{\sqrt{2}+4}{2}, \frac{4-5\sqrt{2}}{2} \right) = (1, \dots) \rightarrow \text{desired: } \left(1, 2 - \frac{5\sqrt{2}}{2}, 2 - \frac{3\sqrt{2}}{2} \right)$$

$$7) \phi \text{ isom. } (\Rightarrow) AA^T = I_3.$$

$$\frac{1}{2} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -3 \\ 0 & -1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & -4 \\ 0 & 2 & -4 \end{bmatrix} \neq I_3 \Rightarrow \boxed{C} \text{ not isom}$$

$$8) \text{ Desired: } \phi\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \frac{\sqrt{2}}{2} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & -1 & -3 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

mine:

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -3 \\ 0 & -3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

$$9) \text{ Ref}_{\Pi}^{\perp}(X) = \frac{1}{9} \begin{bmatrix} -7 & -4 & 4 \\ -4 & 1 & 8 \\ 4 & 8 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = HX$$

$$a_{n+1} = 0.$$

H

~~H~~

! General reflection formula: $\left(I_m - 2 \cdot \frac{a \otimes a}{|a|^2} \right) X - 2 \cdot \frac{a_{n+1}}{|a|^2} \cdot a =$

$$\Rightarrow a_{n+1} = a_4; \text{ plane } \Pi: a_1 x + a_2 y + a_3 z + a_4 = 0$$

$$\Rightarrow a \otimes a = a \cdot a^T = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} [a_1 \ a_2 \ a_3] = \begin{bmatrix} a_1 a_1 & a_1 a_2 & a_1 a_3 \\ a_1 a_2 & a_2 a_2 & a_2 a_3 \\ a_1 a_3 & a_2 a_3 & a_3 a_3 \end{bmatrix}$$

$$|a|^2 = (\sqrt{a_1^2 + a_2^2 + a_3^2})^2 = a_1^2 + a_2^2 + a_3^2$$

$$\text{lin. part} = I_m - 2 \cdot \frac{1}{a_1^2 + a_2^2 + a_3^2} \cdot \begin{bmatrix} a_1 & \dots \end{bmatrix}$$

$$\boxed{R(X) = X} \Rightarrow HX = X \Rightarrow (H - I_3)X = 0.$$

$$\Rightarrow \left(\frac{1}{9} \begin{bmatrix} \dots \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} \right) X = 0$$

$$\Rightarrow \frac{1}{9} \begin{bmatrix} -2 & -4 & 4 \\ -4 & -8 & 8 \\ 4 & 8 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \quad | \cdot 9$$

$$\Rightarrow \begin{cases} -2x - 4y + 4z = 0 \quad |:2 \Rightarrow x + 2y - 2z = 0 \\ -4x - 8y + 8z = 0 \\ 4x + 8y - 8z = 0 \quad |:4 \Rightarrow x + 2y - 2z = 0 \end{cases}$$

1) $\Pi: x + 2y - 2z = 0$ the plane.